



DCA INSTALLATION GUIDE

REV. 15/04/2021
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Together for the environment

MPS Monitor supports the PrintReleaf Programme which measures paper consumption through an aggregation process that uses the automatic transmission of data from the printing devices detected by our system. This allows end users to help neutralize any waste of paper, improve their environmental impact and participate fully in supporting our global forests, in a fully automatic and certified manner. This contributes to reforestation based on a ratio of 8,333 sheets of paper per tree. This equation is specified in PrintReleaf's global standard for sustainability.

Please only print this manual if it is absolutely necessary. Thank you.



About us

MPS Monitor srl is a company headquartered in Italy, with offices in Europe, Asia and Americas. It is a global leader in IoT software solutions for remote printer monitoring and managed print services (MPS). MPS Monitor provides an extremely complete SaaS platform for monitoring and managing printers, available in the Cloud and equipped with a full set of APIs for integrating external software solutions. Its highly scalable and high-performing data collection technology allows dealers, large end users, manufacturers and printer consumables retailers to satisfy any need associated with the management of printing devices, recording of page volumes and supply of toner in an automatic and controlled way.

MPS Monitor Cloud system operates within an ISO/IEC 27001:2014-certified Information Security Management System (ISMS), and guarantees full compliance with the EU General Data Protection Regulation (GDPR).

BuyersLab / Keypoint Intelligence in 2020 assigned to MPS Monitor a Platinum Rating, and certified its compliance to the BLI Security Validation Testing: Policy Compliance. MPS Monitor platform is Certified for HP Smart Device Services 2.0.

MPS Monitor is a member of the BTA - Business Technology Association.

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MPS Monitor S.r.l.

Via dei Borromei, 2 – 20123 Milan - Italy

www.mpsmonitor.com

Tel. +39 02.84.800.612 - help@mpsmonitor.com

VAT Number : 03122040987

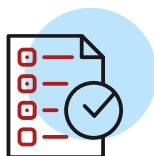
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1

Requirements



- *Hardware requirements (Windows)*
- *Software requirements (Windows)*
- *Network security requirements*
- *Antivirus and firewall*



1.1

Hardware requirements (Windows)

The recommended hardware configuration for the DCA connector for Windows is shown below:

- RAM: 2 GB
- CPU: Intel Core 3
- HD space: at least 100 MB

The PC does not need to be dedicated, and can be used simultaneously for other uses. The DCA connector runs totally in the background and PC users are not affected in any way when using other applications.

The PC must have Internet access via HTTPS and be able to reach the printers to be managed, via the TCP and UDP protocols.



- For installations on small and medium networks (from one to a few dozen printers), configuration with recommended hardware requirements is largely sufficient to scan in seconds or at most in minutes (depending on how many devices there are).
- For large networks, if you have a lot of printers to scan (hundreds or thousands) and require short scanning times, the rule better hardware -> better performance applies.

The DCA connector has a scanning speed proportional to the hardware available to it and the extent to which it is intended to be used. It is possible, on very large networks and with high power hardware, to construct a configuration with high parallelism, which drastically reduces data detection schedule.

1.2

Software requirements (Windows)

The DCA Connector needs to run on a system with one of these Operating Systems:



- PC with operating system based on Windows 10.
- Server with Windows Server 2016 or 2019 installation.

The Microsoft .Net Framework 4.5 Full module must be present on the host Windows system. The DCA 3 installation procedure automatically detects the presence of the .Net Framework 4.5 Full on the system and, where it is missing, redirects the user to the following download page, so that it can be installed: [this link](#).

The installation of the .Net Framework 4.5 Full can be done before or during the DCA installation. The DCA installation cannot be completed without .Net Framework 4.5 Full. At the end of .Net Framework 4.5 Full installation, the system may require a reboot or it the process may be able to complete without the need for a reboot.

We therefore advise you to take into consideration, especially if you are operating on a Windows Server platform that has not recently been updated, and on which there are other services in production, that a system reboot may be required by Windows at the end of .Net Framework 4.5 Full installation.

The installation of the DCA does not require the installation or configuration of any other particular software component (e.g. Database, Web server, etc.), as all data management and web application functions are delegated to the Portal.

1.3 Network security requirements

The DCA connector must be hosted on a system that has stable access via HTTPS (port 443 outgoing), to the following domains:

- https://*.mpsmonitor.com
- https://*.abassetmanagement.com
- <https://jamanagement.hp.com>
- <https://jamanagement.api.hp.com>



- <https://connectivity.pod1.avatar.ext.hp.com:443/avatar/v1/entities/connectivityconfig>
- <https://ews.hpjamservices.com>
- <https://registration.pod1.avatar.ext.hp.com:443/avatar/v1/entities/credentials>
- <http://sd.symcb.com/sd.crl>
- <http://svrsecure-g3-crl.verisign.com/SVRSecureG3.crl>
- <http://crl.sca1b.amazontrust.com/sca1b.crl>
- <http://crl3.digicert.com/DigiCertGlobalG2TLSRSASHA2562020CA1.crl>
- <http://crl4.digicert.com/DigiCertGlobalG2TLSRSASHA2562020CA1.crl>
- <https://eu.jamanagement.hp.com>
- <https://jamanagement.eu.api.hp.com>
- <https://eu.ews.hpjamservices.com>



Please note that, depending on the country where the DCA is installed, the access to some of the listed domains may not be needed.

If the Internet access via HTTPS is managed through a Proxy Server:

- If the Proxy does not require a specific authentication, the connector does not require configuration, as it is able to recognize and interface with the Proxy automatically.
- If the Proxy requires authentication credentials, a Proxy user is required that is enabled to reach the domains listed above.



Any specific proxy user will be inserted during installation (or subsequent configuration) of the connector.

The credentials (user name and password) for HTTPS access via Proxy are saved by the connector, in encrypted form, only and exclusively on the Customer's system; they are never transmitted to the Portal or to other sites outside the Customer's network, and are never visible in plain text to personnel accessing the Customer's system.

Within the customer's internal network, the following local ports must enable the communication between the PC or Server where the DCA connector is installed, and the IP addresses of the printers:

Port	Protocol	Port	Protocol
161	UDP	3910	TCP
427	UDP	3911	TCP
443	TCP	7627	TCP
3702	UDP	8080	TCP

The connector installation procedure installs several Windows system services on the host system, which are automatically configured and by default use the Local System user to perform all its tasks.

If the network and the host system have no specific restrictions on the Local System user, no change to the default configuration is required, nor is there any additional requirement.

If, on the other hand, there are specific restrictions or policies that prevent the Local System user from activating or executing the service, a specific local user will be needed.

The connector installation procedure will assign this user the minimum rights necessary to perform the service.

Please make sure that the customer's firewall and security infrastructure enables running the application and reaching the needed domains for your country.



In case of need, contact our Support team for help on identifying the right security configuration to enable DCA operations in a specific customer.

1.4 Antivirus and firewall

Antivirus : is a software that can block the execution of certain programs installed on the PC or Server in order to limit the entry of malware or malicious codes into the system that may damage it.

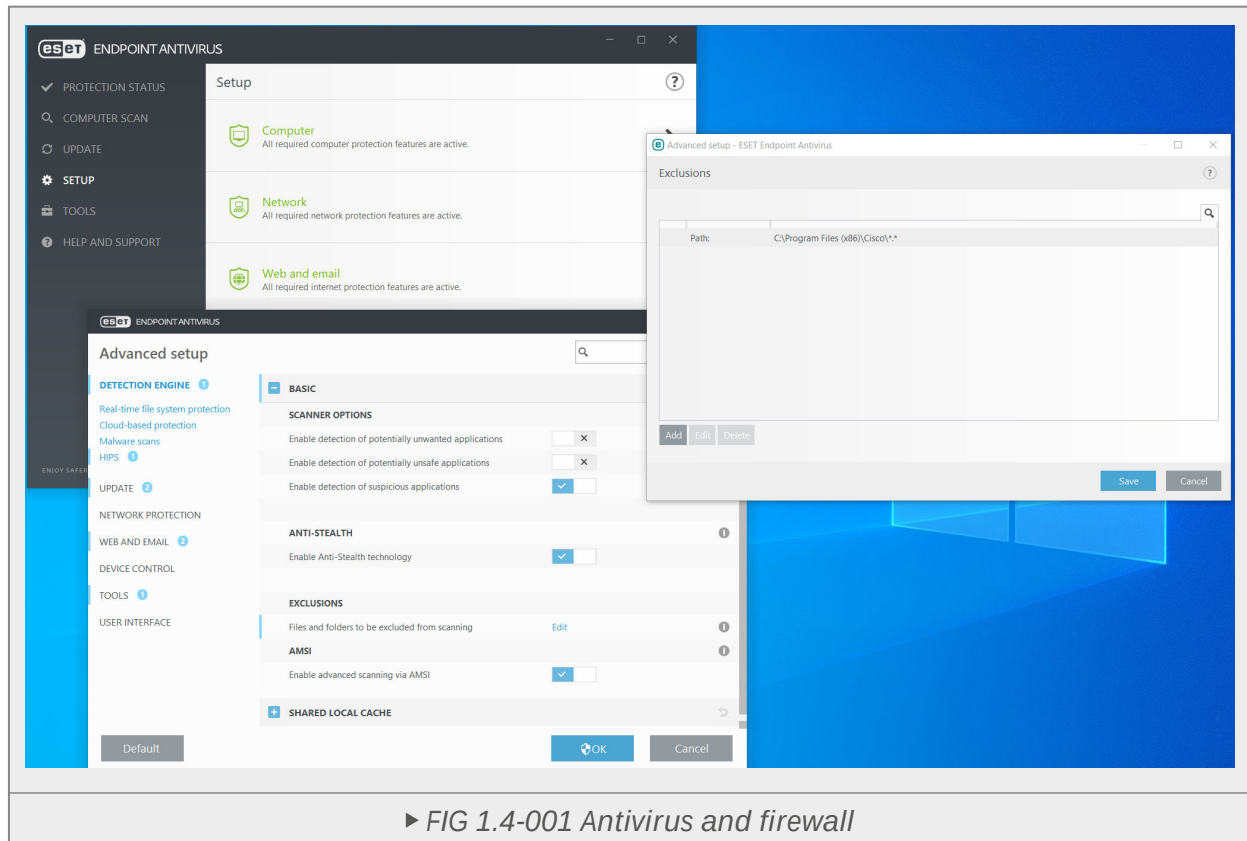
Some antivirus software may recognize system's DCA connector as a threat and proceed to block it. To allow the DCA connector to communicate the data detected by the printers and store them on the cloud, the path in which the DCA connector was installed must be included in the exclusion list of the antivirus, in this case the paths that must be excluded from the antivirus block are: C:\Program Files (x86)\ABAsset or C:\Program Files (x86)\Mpsmonitor (for new installations) and C:\Program Files (x86)\HP JetAdvantage Management. These are the standard paths on which the software is installed during automatic installation. To avoid a blockage of the DCA Connector, you will have to exclude these files as well:

MpsMonitor.eXplorer.ConfigurationUtility.exe and MpsMonitor.eXplorer.Service.exe - Agent\SNMPAgent.exe.

If you chose a different path, then that path must be entered.



Below is a screenshot of the exclusions from the antivirus blocker taken from one of the most common pieces of software in this category:



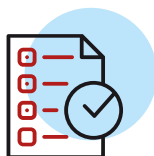
► FIG 1.4-001 Antivirus and firewall

Firewall: is a perimeter defence component of a computer network that acts as a link between two or more network segments, thus providing security protection to the network itself.

This service is already available in operating systems such as Windows and enables outgoing or incoming traffic on certain ports or services within a company or home network, for example, to be blocked.



2 Installation



→ *DCA Connector installation*



2.1 DCA Connector installation

The DCA Connector sits between the various components of the system. The DCA Connector is installed on the PC or server in the customer's LAN and performs reading operations following the instructions provided by the server component on the Portal. The information is transmitted to the Server components using only the secure HTTPS protocol.

2.1.1 Installing DCA for Windows



CAUTION

The custom installation package should only be used for the customer to whom it refers, to allow the system to store the data correctly in the archive. In the event of multiple installations or should you need to reinstall the package, it must be downloaded from the site at the time of use and should not be saved and used afterwards.

If you create a customer, you can download a custom installation package for that customer. During the installation procedure, you will see the customer's own code/description: always make sure to install the corresponding package for the customer concerned.



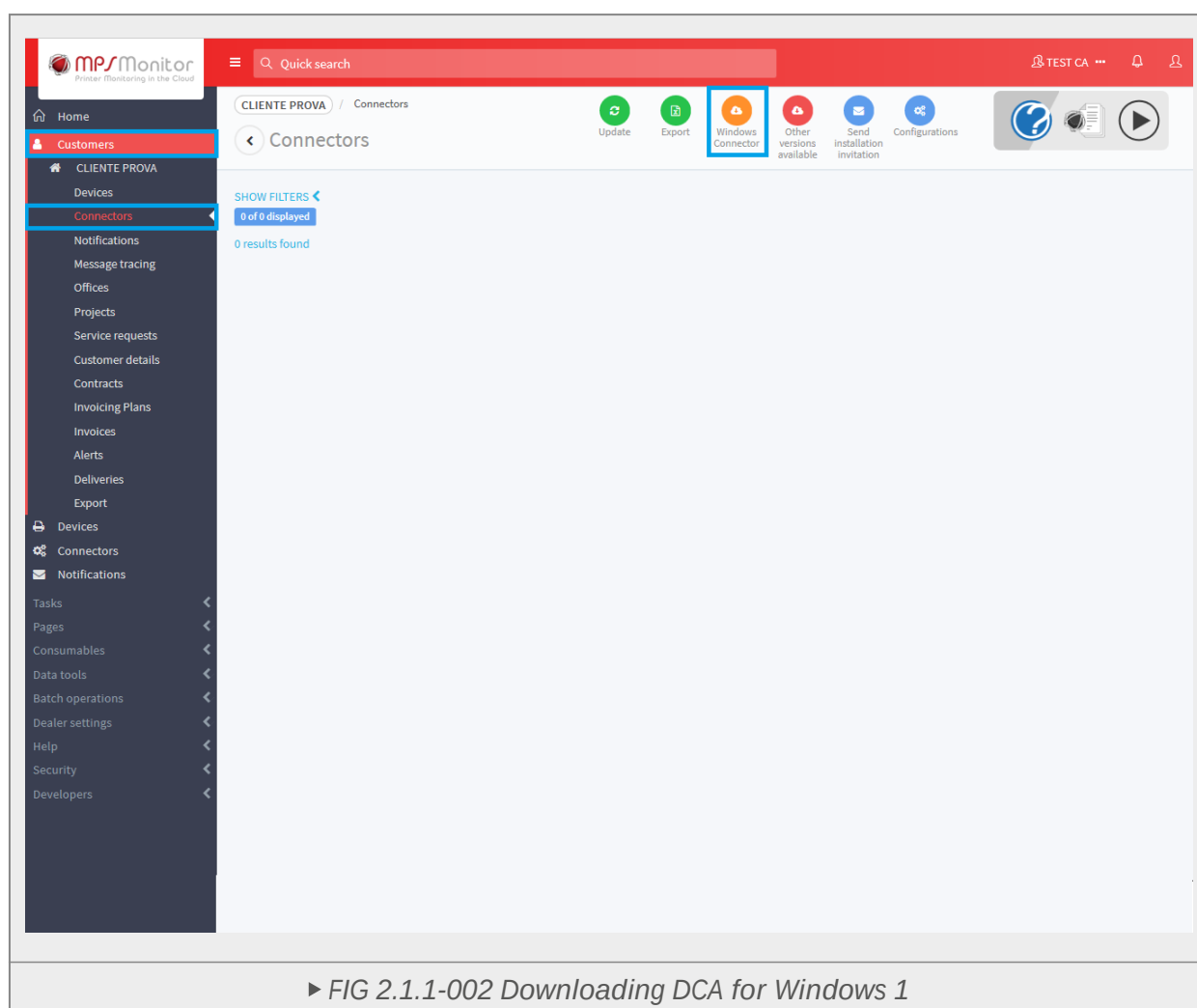
For all other connectors and operating systems, you can consult the [2.1.2 Installing the connector for all versions other than Windows](#) section.

To install the DCA Connector on the customer's computer, you will need direct access to the customer's computer, either via virtual machine or with the help of remote access software.



2.1.1.1 Downloading DCA for Windows

To download the DCA for windows connector you have to access the Customer connectors section: on the MPS Monitor Home Page, select Customer -> Connector, then select the windows connector button in the upper part of the window, as shown in the figure below.





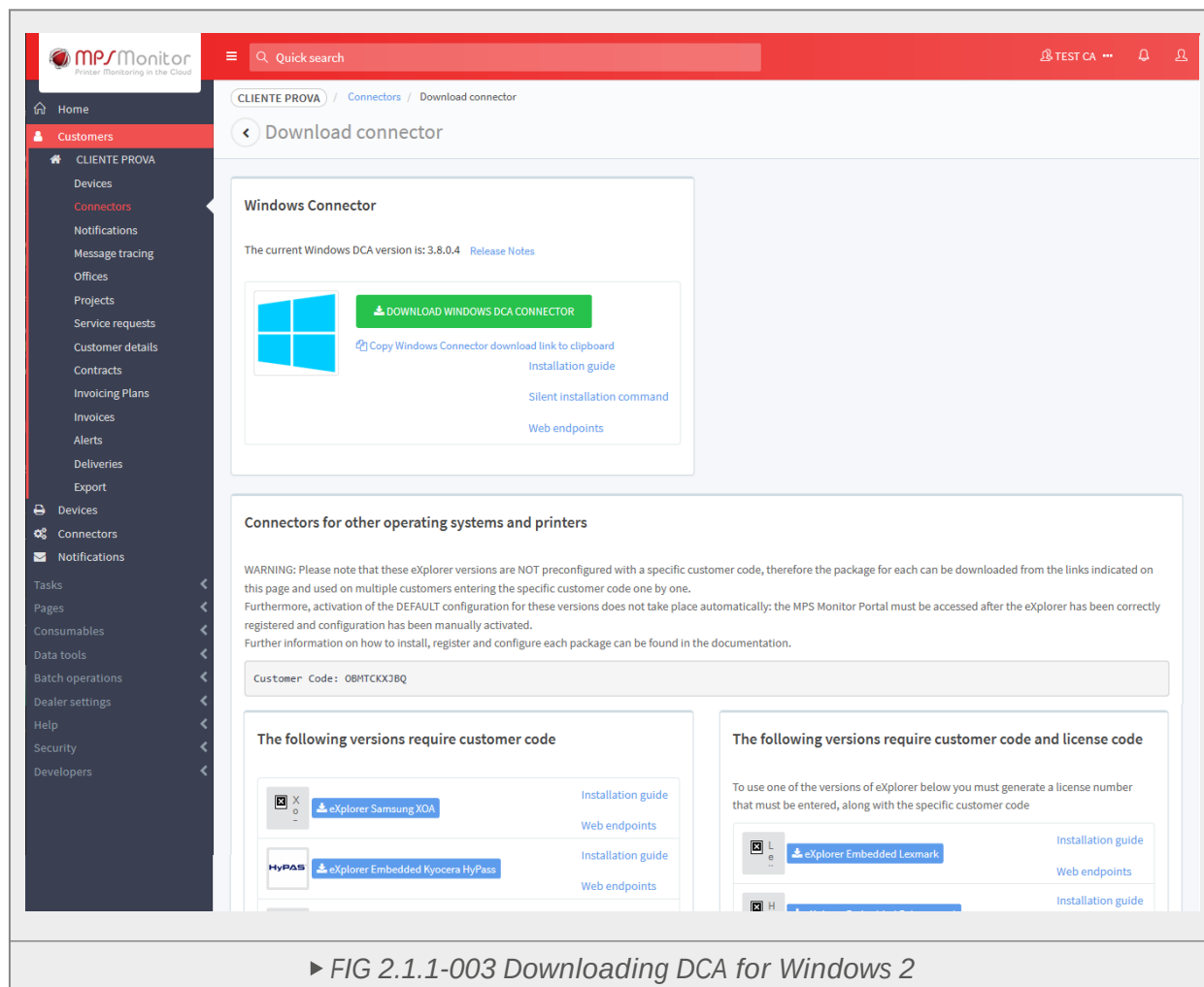
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mpsMonitor
Printer Monitoring in the Cloud

The download connector page opens.

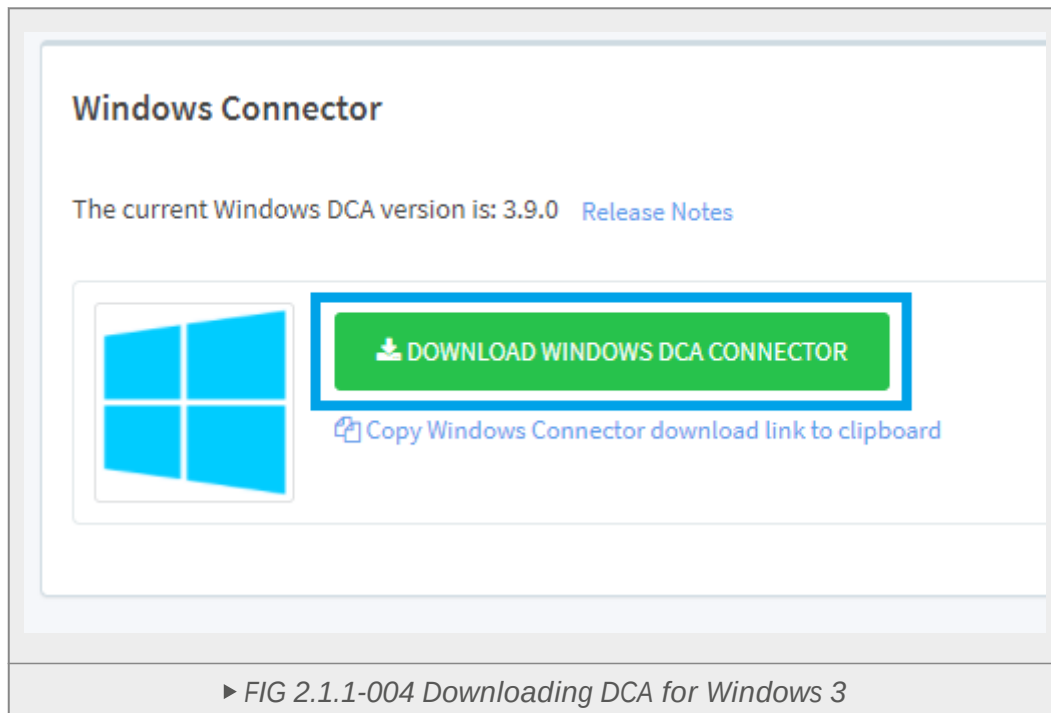


► FIG 2.1.1-003 Downloading DCA for Windows 2

The MPS Monitor provides, for the versions 3.9.0 and higher of the DCA Connector, a page viewing the data about the download package. It allows to see clearly the name of the Customer to whom the package is referred and to avoid errors caused by downloading a package for customer different than the one for whom the download is in progress.

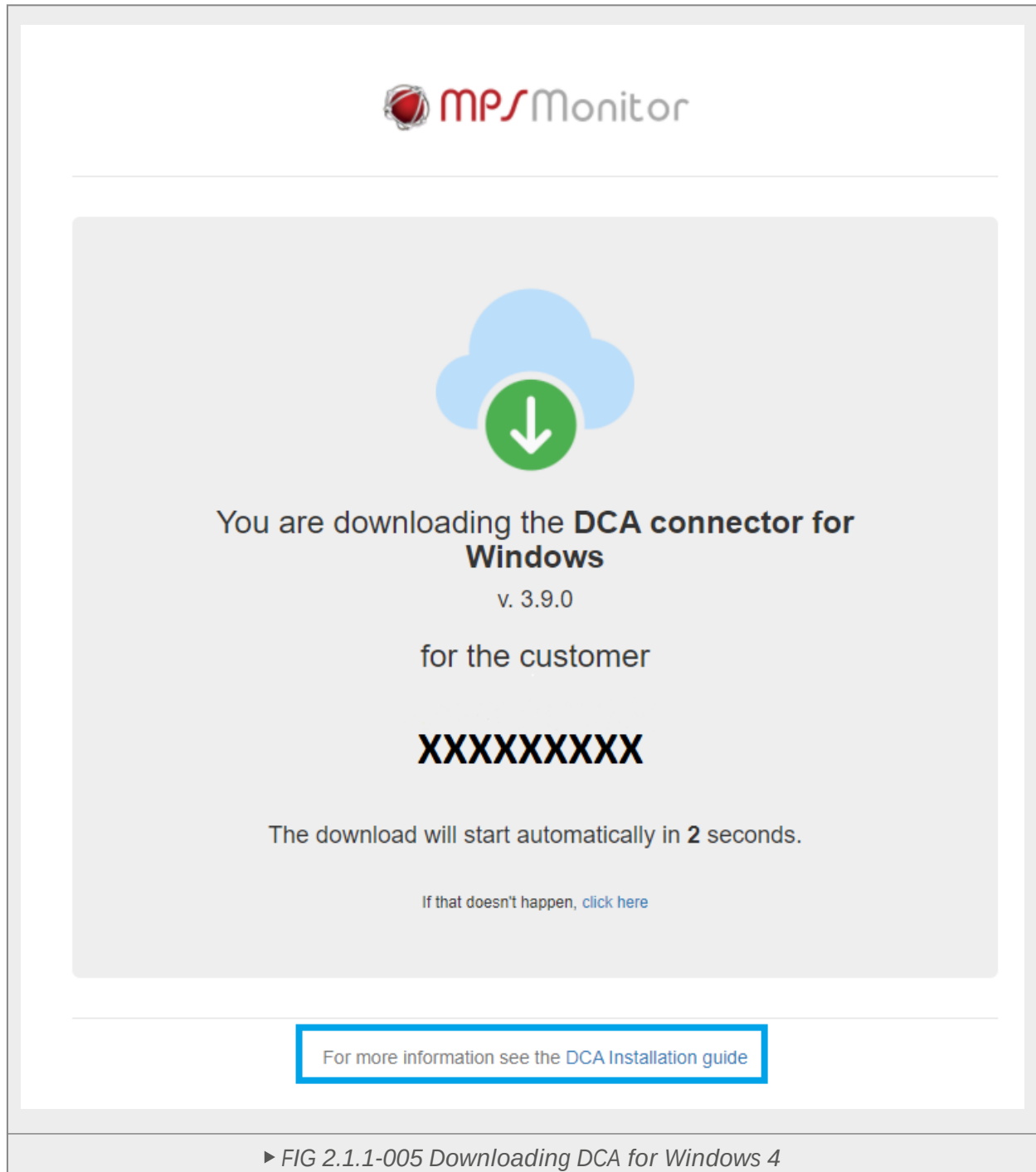


Select the download windows DCA connector button to start the download.





The page with DCA Connector version number, Customer's name and Download countdown will open.





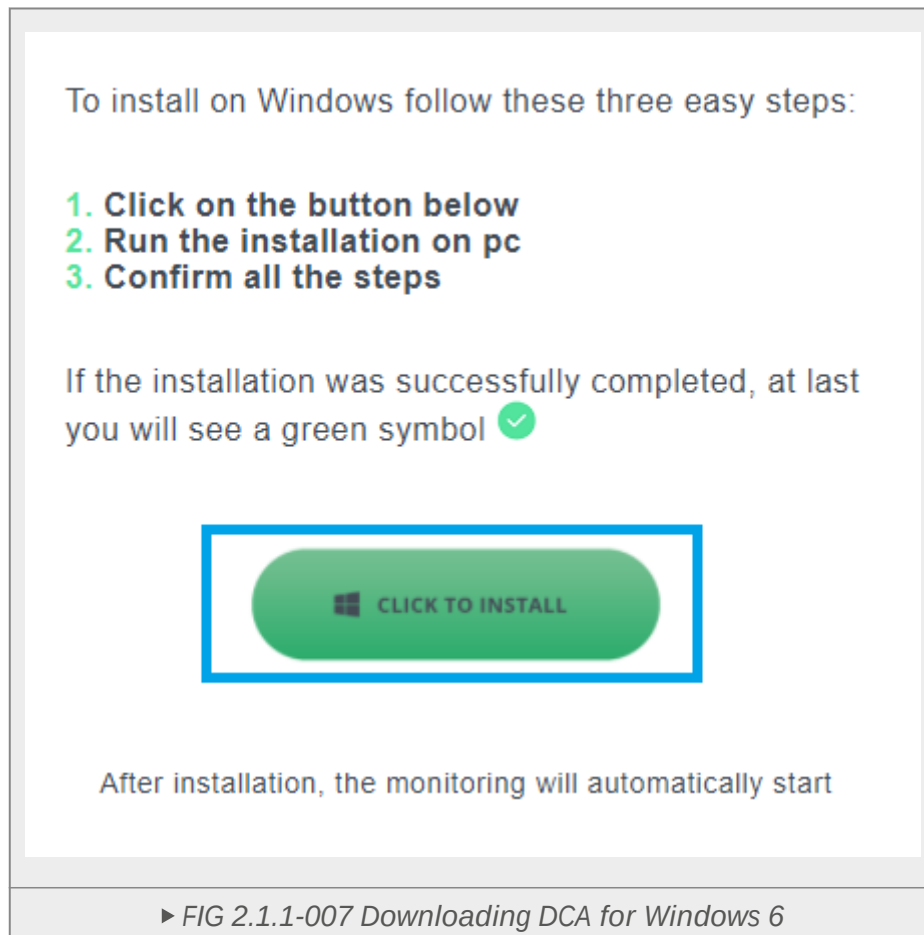
The download procedure is performed in the same way if:

- you copy the link from the Portal and send it via e-mail selecting Copy Windows Connector download link to clipboard.



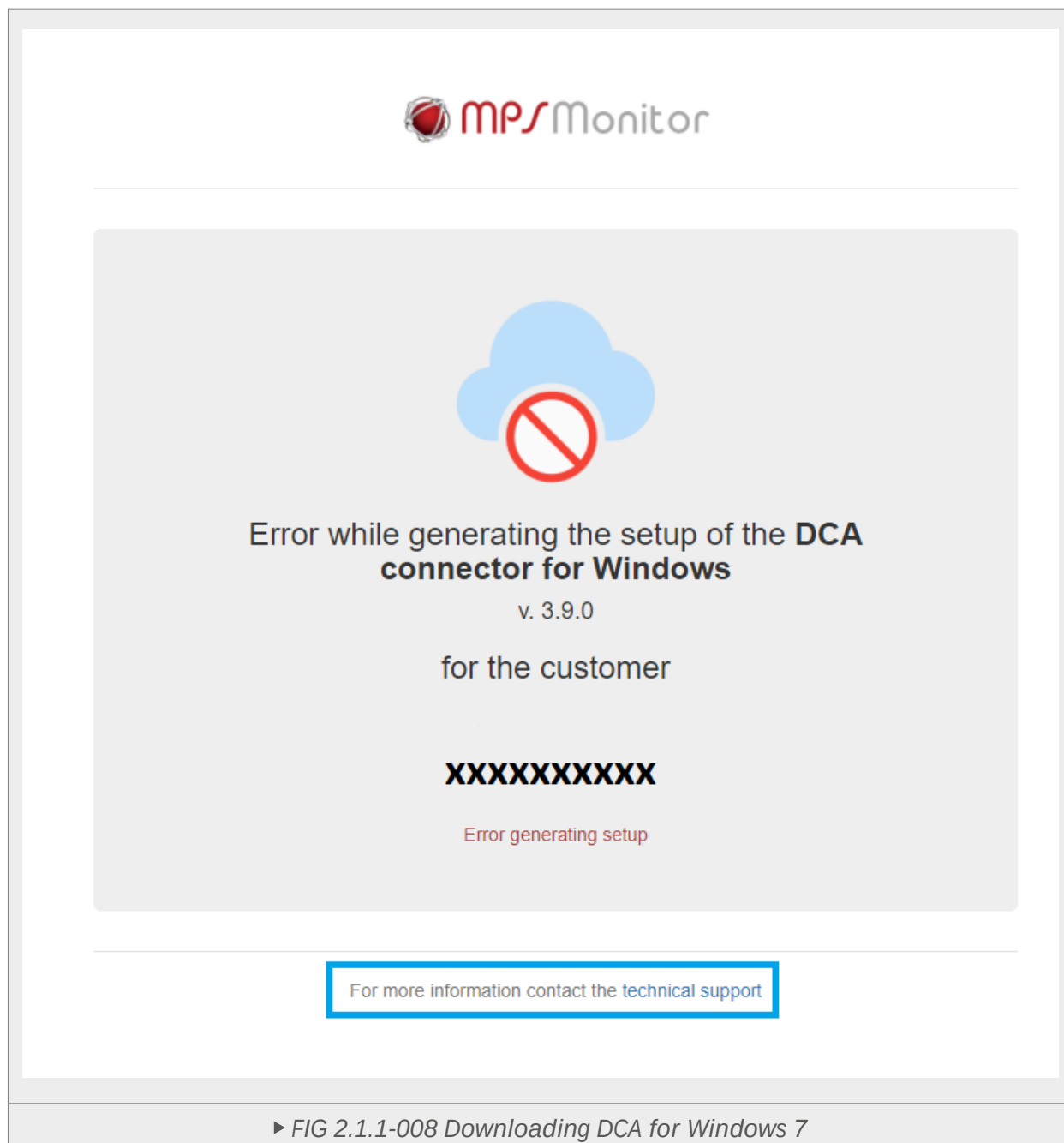


- the installation is started from the link in the installation e-mail received selecting the Click to install button.





In case the download procedure fails to generate the Setup package, the following message will be displayed.



► FIG 2.1.1-008 Downloading DCA for Windows 7



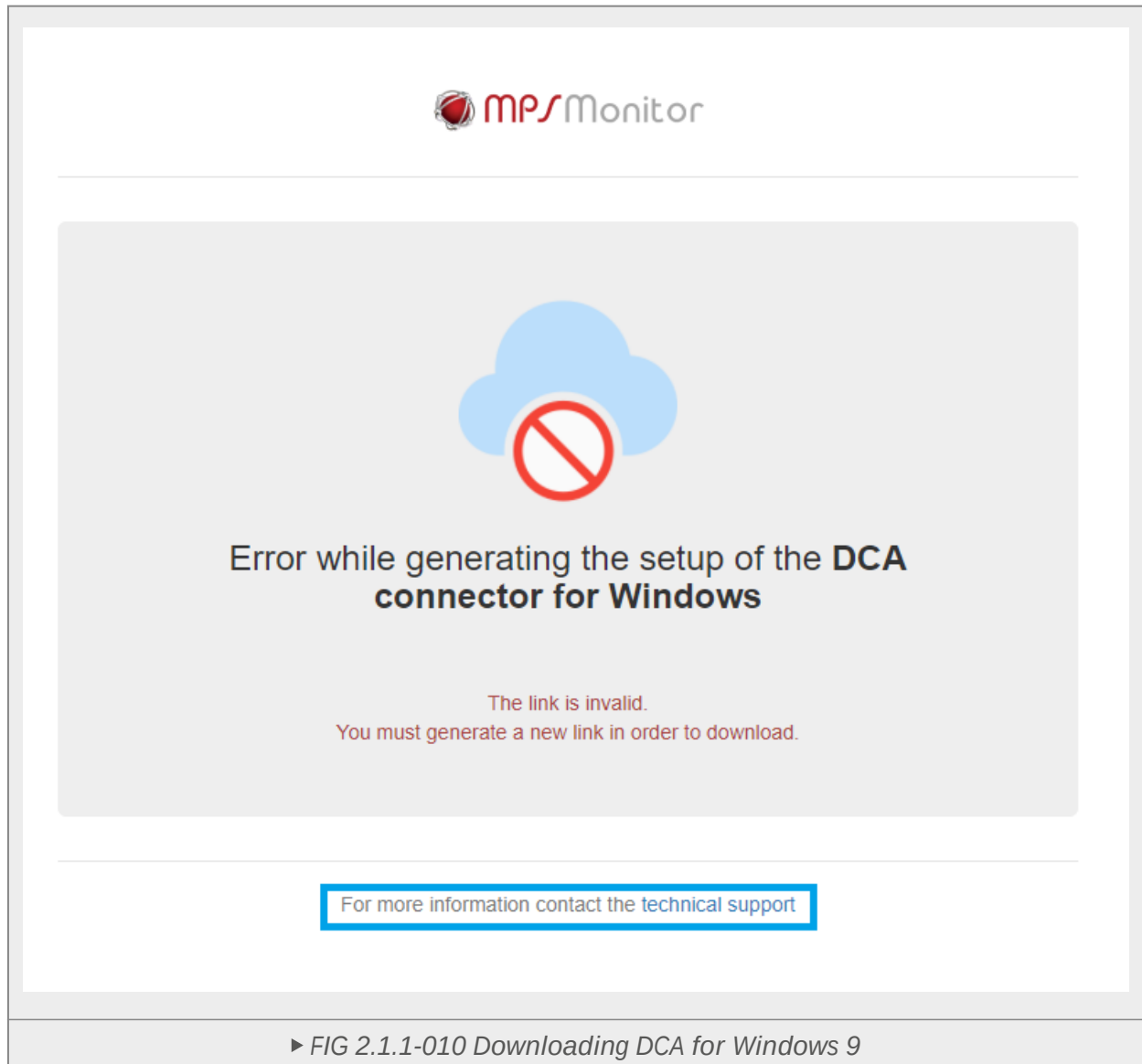
In the footer there is the link that allows to generate a support request via e-mail to help@mpsmonitor.com. It is already precompiled with the subject that allows to identify the problem immediately.

To	help@mpsmonitor.com
Cc	
Subject	Windows DCA Installation Problem - Customer XXX XXX - Dealer MPS Monitor srl - Token 71ce64c9-42a5-4e54-8391-feceb510310b

► FIG 2.1.1-009 Downloading DCA for Windows 8



In case the download link is not valid (e.g. it was not copied correctly), the following message will be displayed.





In the footer there is the link that allows to generate a support request via e-mail to help@mpsmonitor.com. It is already precompiled with the subject that allows to identify the problem immediately.

To	help@mpsmonitor.com
Cc	
Subject	Windows DCA Installation Problem - Token 292045bc-c265-421c-939f-4818093ca96n

► FIG 2.1.1-011 Downloading DCA for Windows 10

The installation procedure for the DCA connector is the same as for the previous versions of the package and described hereafter.

2.1.1.2 Installation execution

To install the DCA Connector, perform the following steps:

- 1 Download the DCA Connector setup from the link provided after creating the DCA Connector Customer position on the MPS Monitor Portal. The link can be generated only by a user with Administrator or Installer credentials on the MPS Monitor Portal. For more information on how to create the link, see the Configuration Guide and Technical Reference for the DCA Connector, or contact your supplier of MPS Monitor system to request it.
or download the DCA connector setup following the procedure *Downloading DCA for Windows* described above.

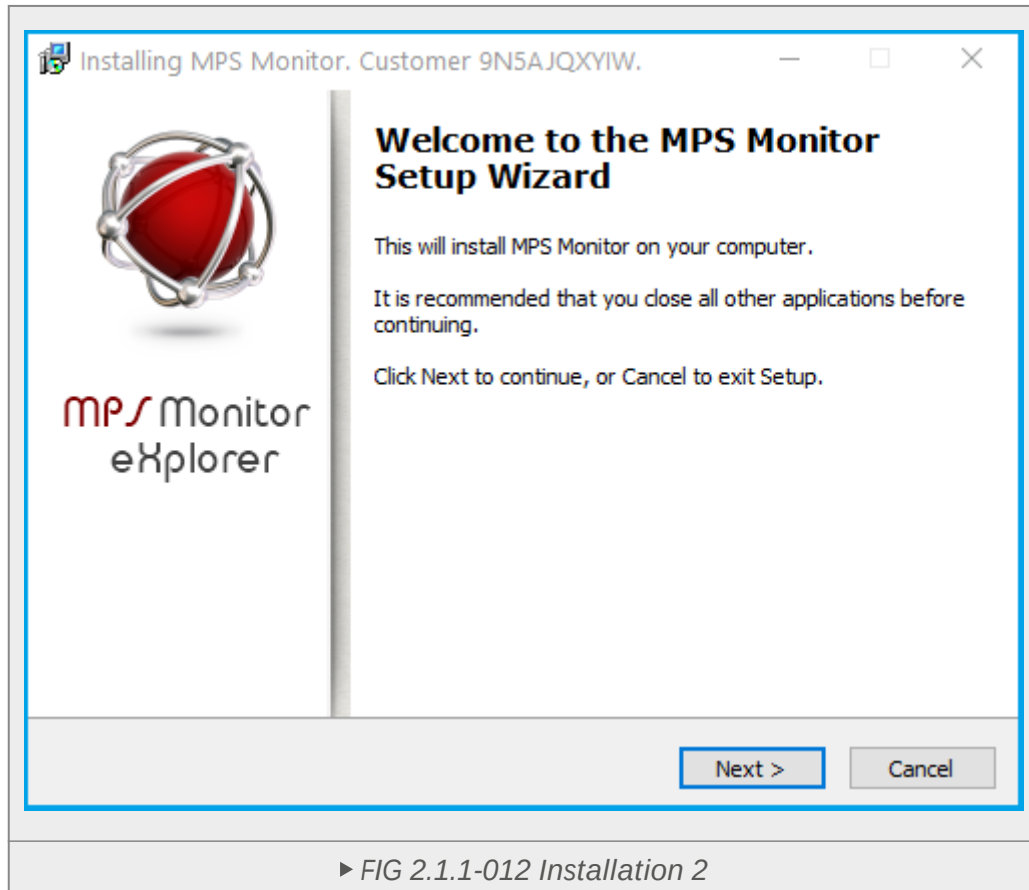


- 2 Extract from the downloaded package the file MpsMonitor.DCAConnector.Setup.exe and run it as administrator. The setup is digitally signed when you download the zip. Then, with a double click on the exe setup will appear this message: Do you want to allow the following program to make changes to this computer? Click Yes



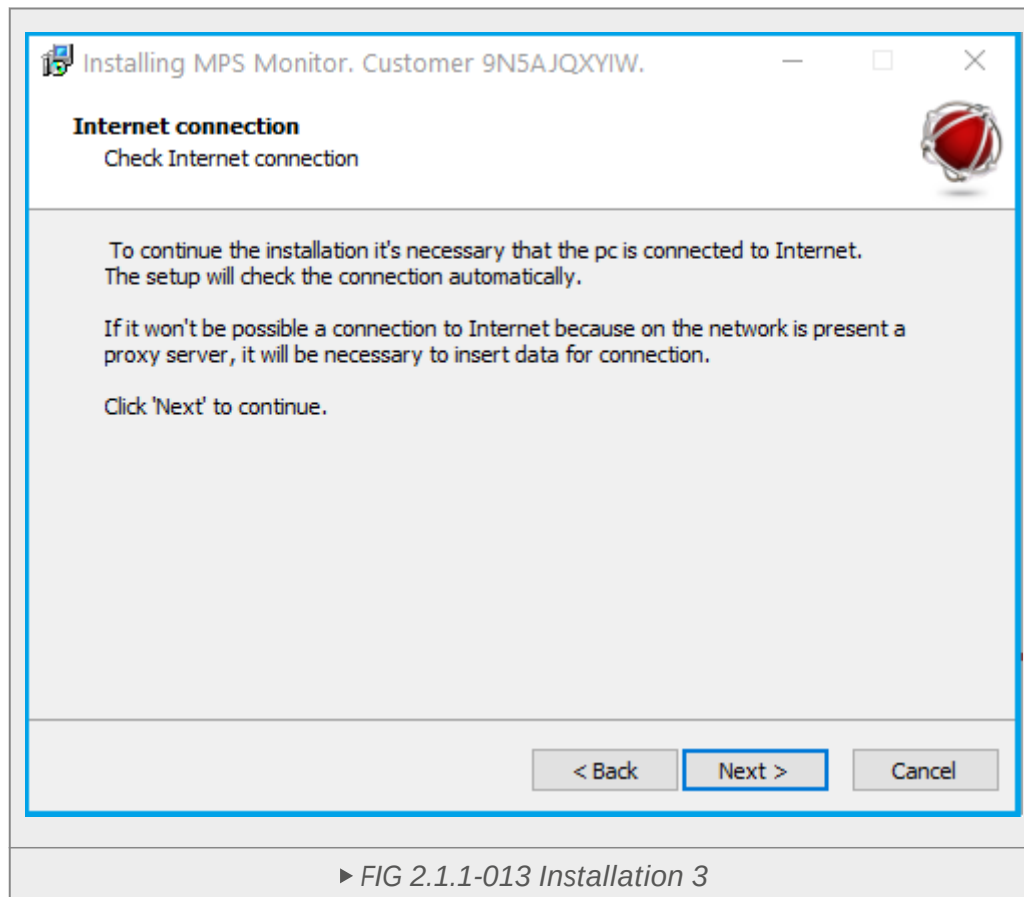
CAUTION

Always check the correctness of the identification number and description of the Customer. If you are using a package that does not correspond to the Customer you want to install, cancel the installation and get the link to download the installation package properly:





- The DCA Connector setup verifies that the PC on which you are installing is connected to Internet and have access to the URL https://*.mpsmonitor.com. Click Next:

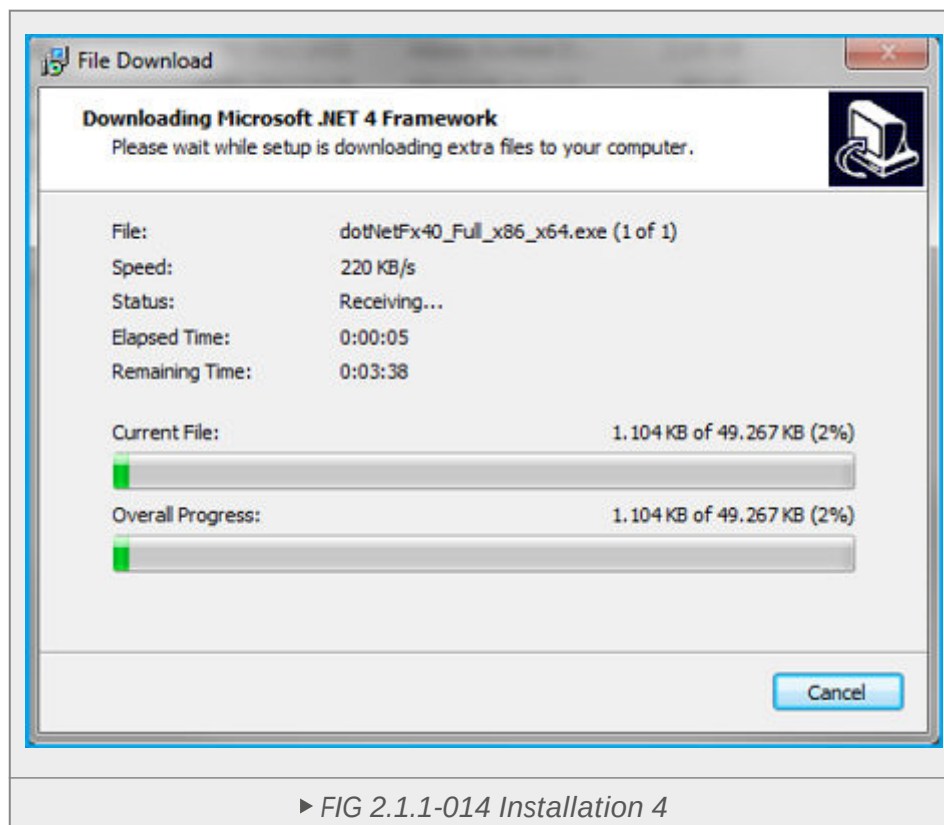


► FIG 2.1.1-013 Installation 3

- If on the hosting system is not already installed Microsoft.Net Framework vers. 4.5 Full, the program prompts a prerequisites request.



Clicking Next, the .NET Framework 4.5 Full will be downloaded automatically from Microsoft site, and like so the subsequent installation, which will be performed by confirming all steps required.



If you required the system restart, you can reboot it at the end of the DCA Connector installation procedure, or reboot it immediately and then redo, from the beginning, the installation process.

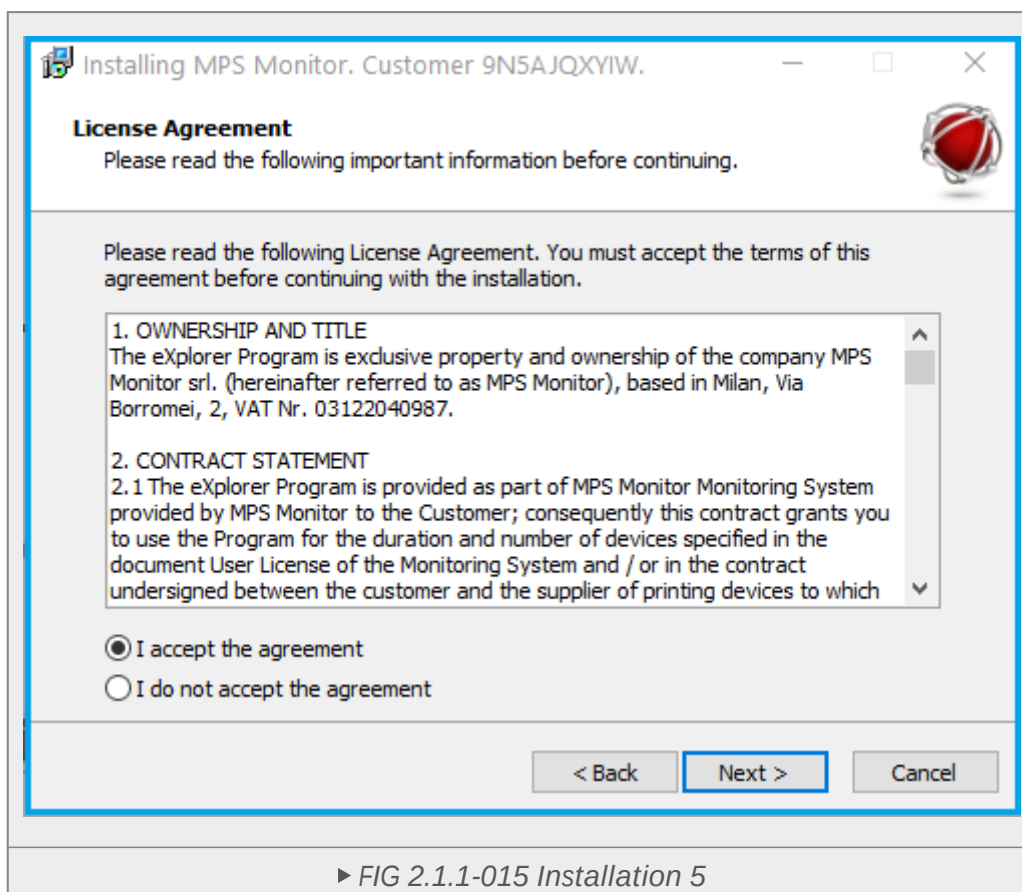


CAUTION

If the system asks for a restart, it must absolutely be done; otherwise DCA Connector will not work until the next system reboot.

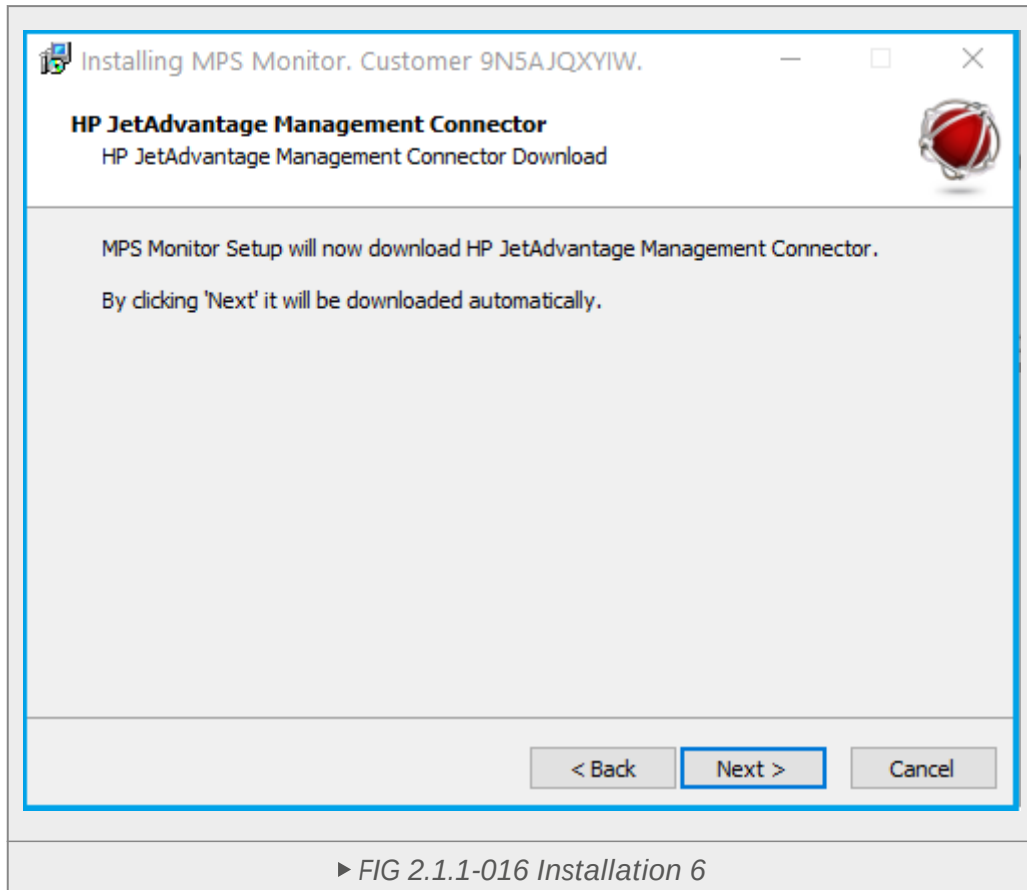


- 5 Accept the license terms and click Next:



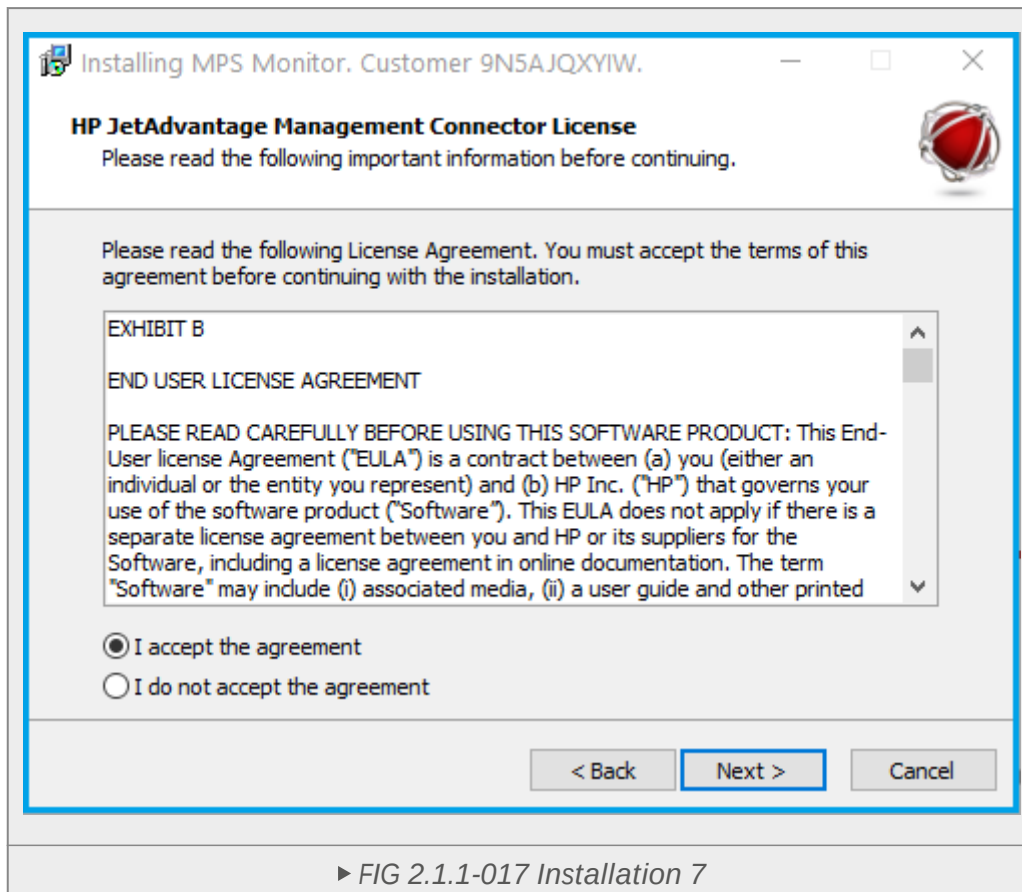


6 Click on Next:



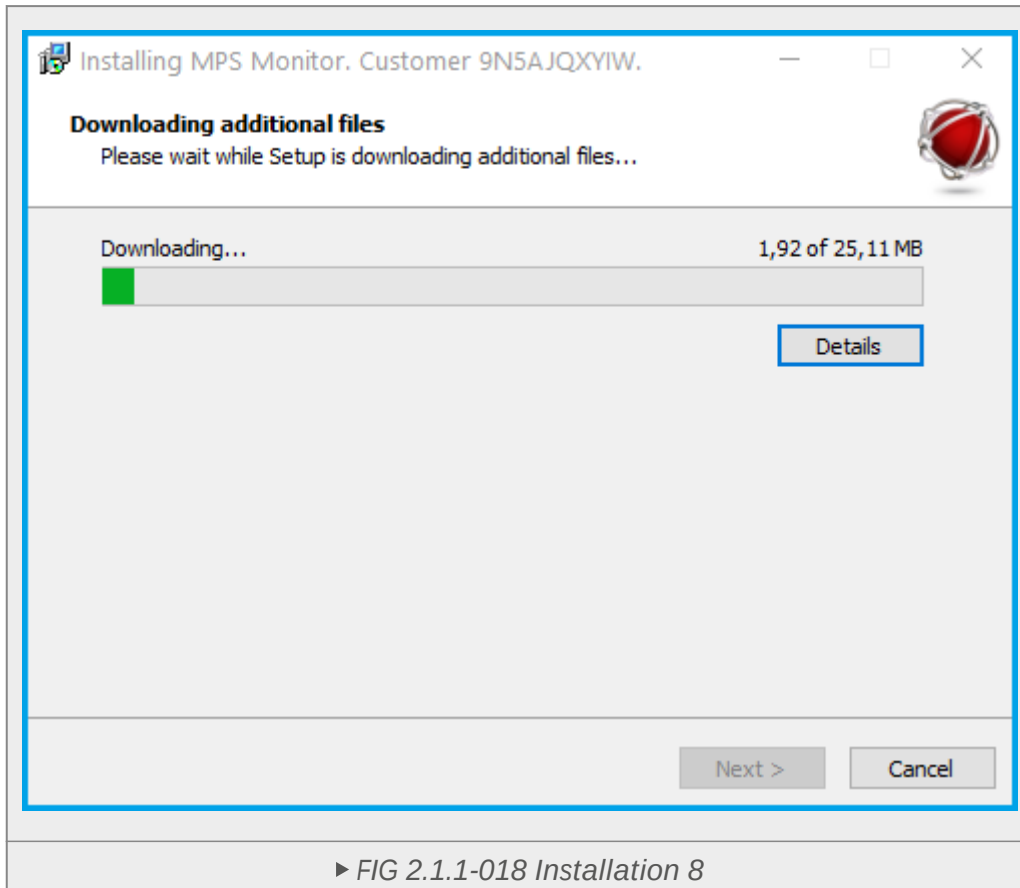


- 7 Accept the agreement and click next:



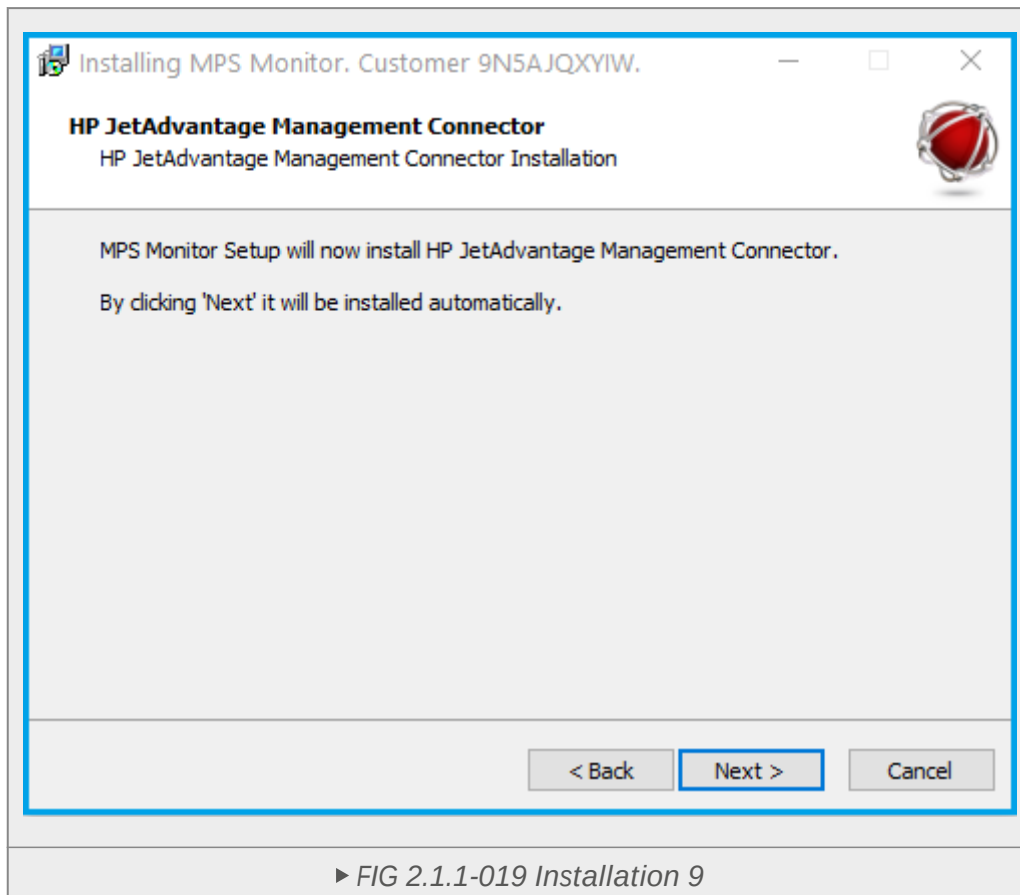


- 8 Wait for the download of additional files:



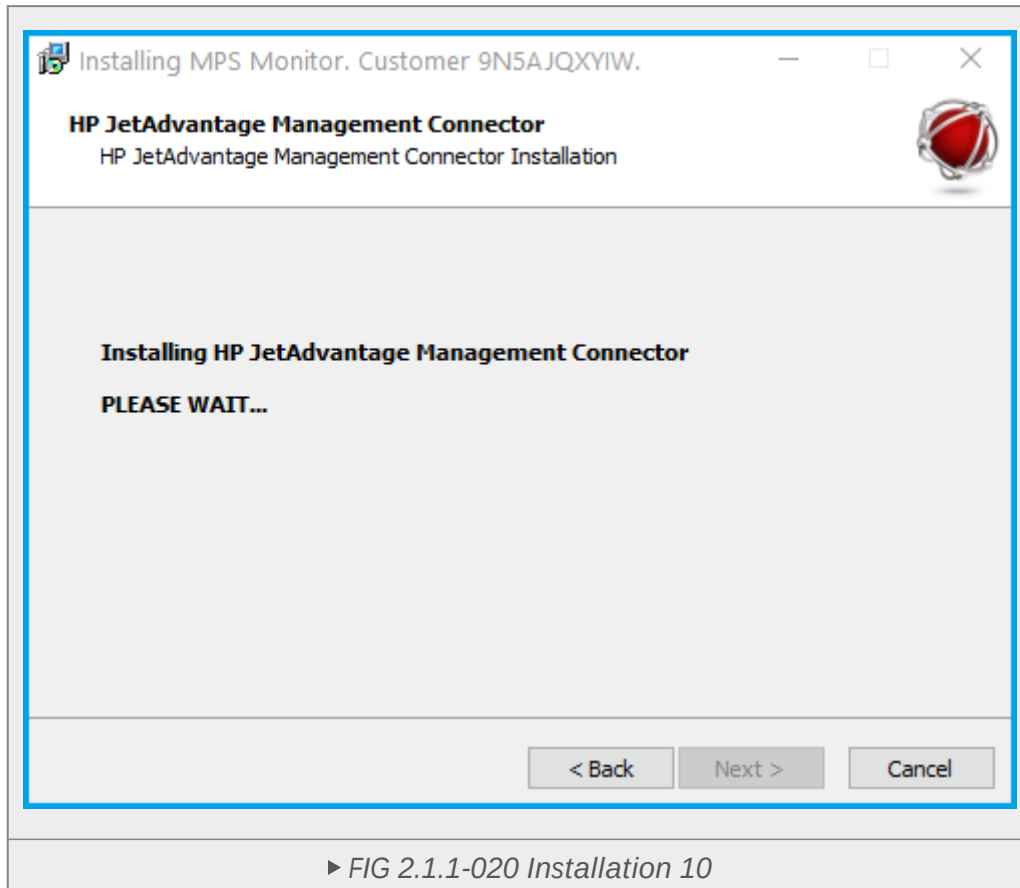


9 Click on Next:





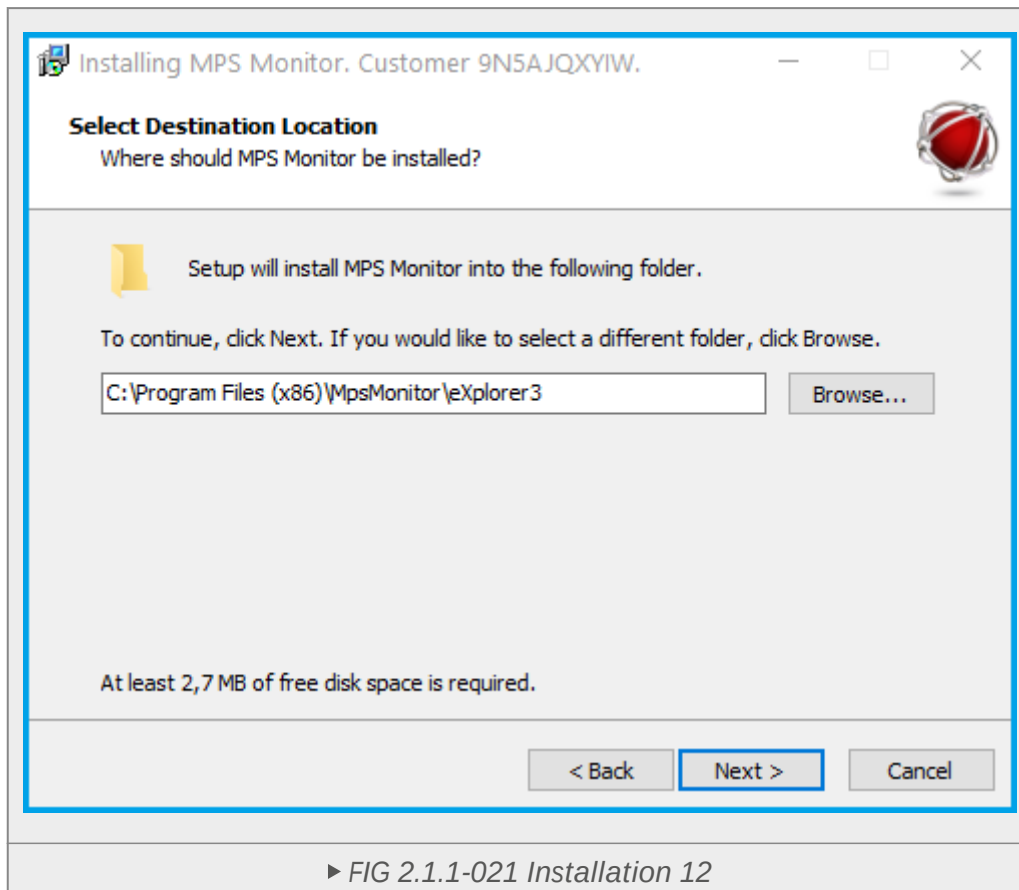
- 10 Wait for the HP Connector installation and then click on "Next":



- 11 Click on next in the following step:

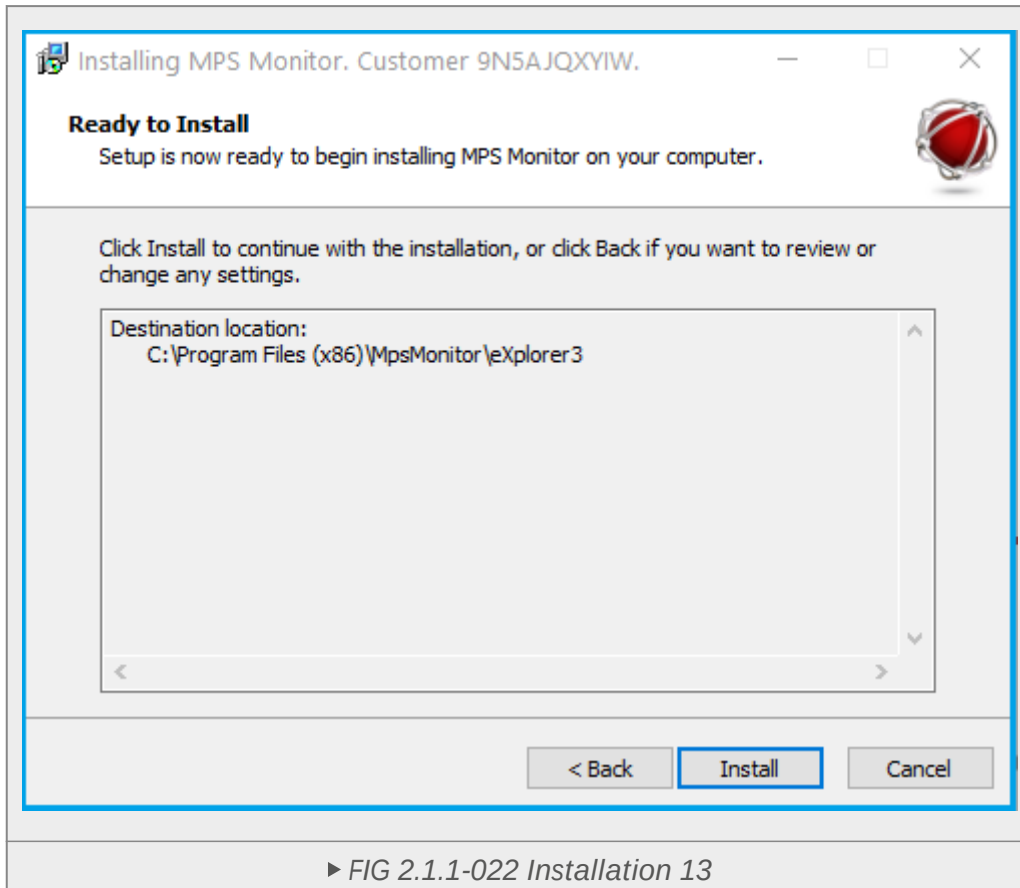


- 12 Check the installation procedure and click Next:





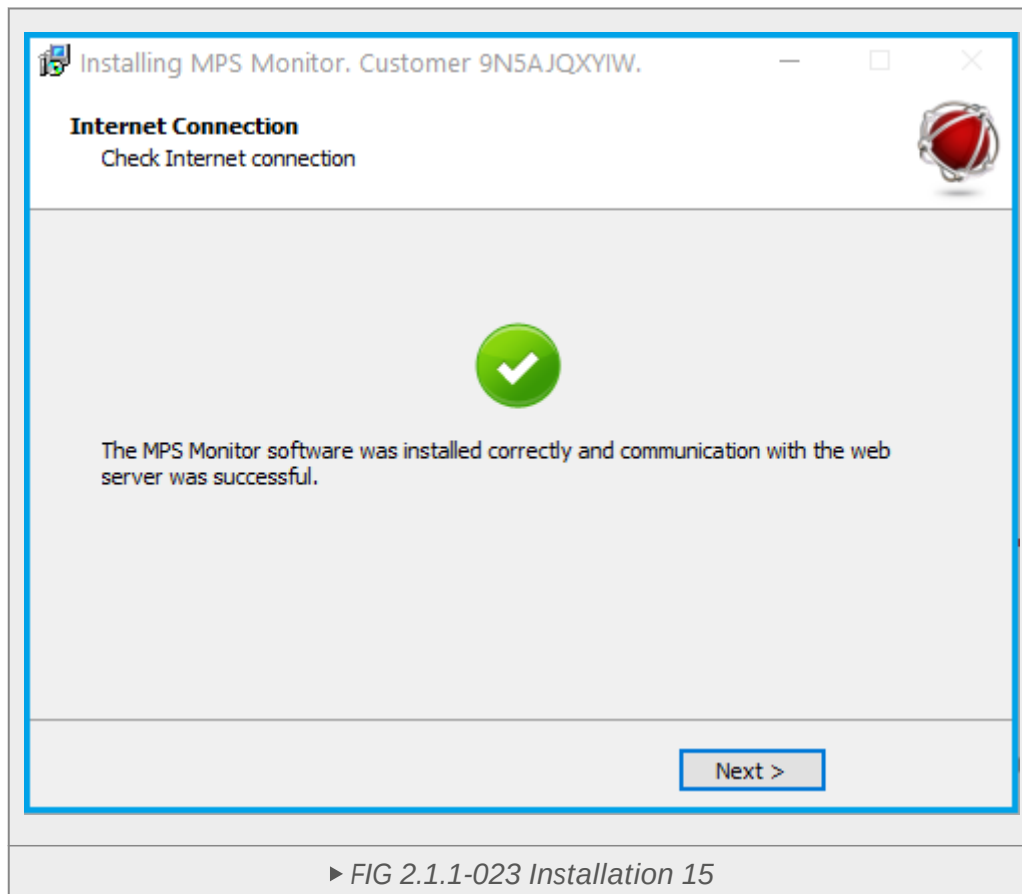
13 Click Next:



14 Wait until the installation is complete:



- 15 If the following window appears, means that the communication control with the web server was successful. Click Next to go directly to step 15 and conclude the installation:



► FIG 2.1.1-023 Installation 15

- 16 If communication with the MPS Monitor Portal is not successful, and it's necessary to configure the DCA Connector startup parameters, click Next to display the configuration screen. At this point it's necessary to enter the required data.

The user of the service MpsMonitor.DCAConnector.Service, by default is Local System, and you don't need to change it unless there are no particular restrictions by the Customer. Once assigned a different account than SYSTEM for the service execution, the setup will automatically assign it the necessary permissions for the correct operation, therefore the account entered here can also be simply a "local user", or in case of domain a "domain user".



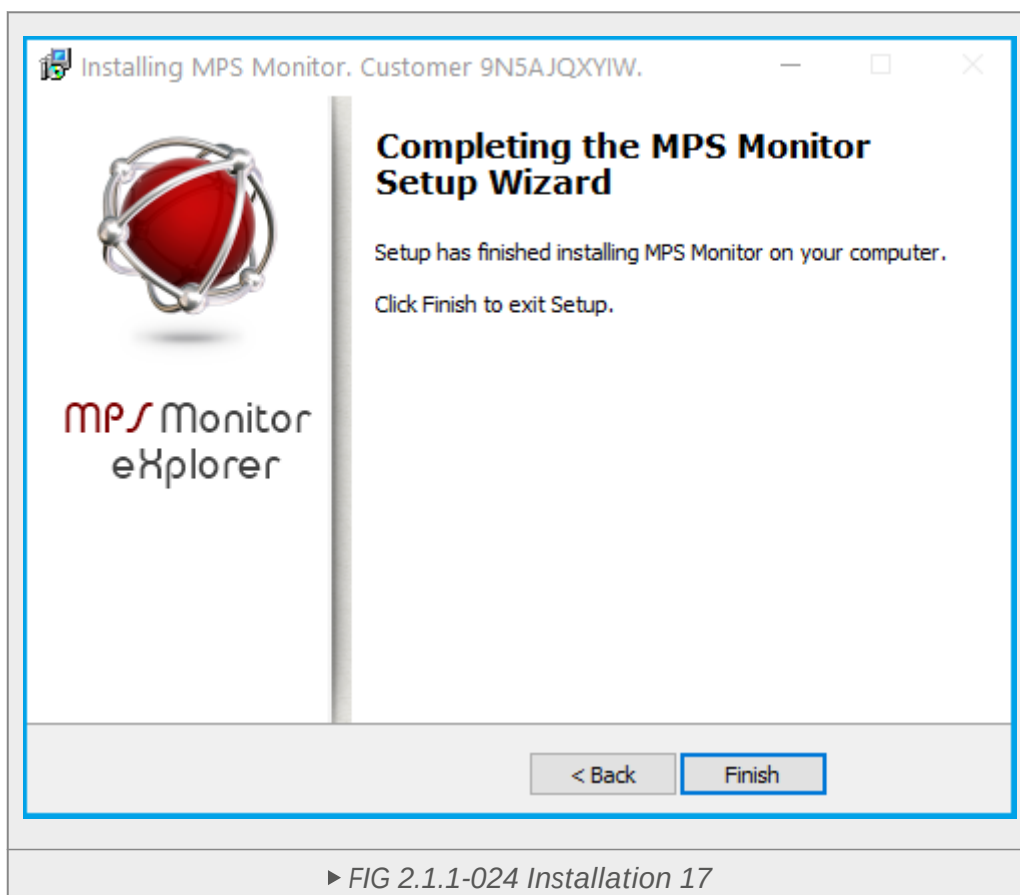
Once entering parameters and press "Apply settings", the procedure will automatically check if, with the parameters entered, the communication with the MPS Monitor Portal is active. Then will appear green or red icons, depending on the check outcome.

The validation of parameters here entered is the verification of the correct installation.



The proxy parameters validation can be successful only if it has already been a successful parameters validation of the user which executes the service.

17 Installation concluded. Click Finish:





2.1.1.3 DCA Connector Auto-Update

Automatic Update from DCAConnector.x to higher versions of the DCA Connector

The update from one version of the DCA Connector to a higher one is performed automatically, with no need to access to the Customer's system, as soon as a new version is published.

If you wish to disable automatic updates on the DCA Connector, please contact technical support at help@mpsmonitor.com.

2.1.2 Installing the connector for all versions other than Windows

In the [2.1.1 Installing DCA for Windows](#) section, you have seen how to install the DCA Connector for the Windows operating system.



Instructions for other versions of the DCA connector are available on a page of the MPS Monitor portal that is easily accessible by following the Customer -> Connectors -> Other versions path, as shown in the image below.

Connectors

The connector installed on this client is not communicating. Check if the customer system is active and eventually reinstall the connector.

SHOW FILTERS < 1 of 1 displayed

System Name: DESKTOP-01SUQ31 IP Address: 192.168.178.109 Last upload: 31/07/2020 14:09:09

Identifier: 3f267cf7-3501-4674-ab5a-7dc9cf89037

Connector		Configurations		
System Name	DESKTOP-01SUQ31	Name	Enabled	Valid
IP Address	192.168.178.109	DEFAULT	Yes	Yes
Last Contacted	13/09/2020 21:40:10			
DCA version	3.7.11.0			
Automatic update	Yes			

System information

Previous 1 Next

► FIG 2.1.2-025 Installing the connector for all versions other than Windows-1

Manuals for the respective DCA connector versions are available at the following links:

- eXplorer Linux Debian-Ubuntu: [click here to access the instructions.](#)
- Samsung XOA explorer: [click here to access the instructions.](#)



- eXplorer Embedded Raspberry Linux: [click here to access the instructions](#).
- Embedded Kyocera HyPass eXplorer: [click here to access the instructions](#).
- eXplorer Mac Os v.2: [click here to access the instructions](#).
- eXplorer Mac Os v.3: [click here to access the instructions](#).

All DCA connectors need the Customer Code, which you can find at the top of the Download other versions of eXplorer page (see next image), which you can reach by following the Customer -> Connectors -> Other versions path.

There are also connectors that, in addition to the aforementioned customer code, require a license code, which is a number that must be generated and entered. To generate a license



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code, you have to click on Generate license, in the bottom right of the page Download eXplorer other versions.

The screenshot displays the MPS Monitor web interface. On the left is a dark sidebar with navigation links: Home, Customers, TEST SDS, Devices, Connectors, Notifications, Print Security, Message tracing, Offices, Projects, Service requests, Customer details, Contracts, Invoicing Plans, Invoices, Alerts, Deliveries, Export, Devices, Connectors, Notifications, Tasks, Pages, Consumables, Data tools, Batch operations, Dealer settings, Help, Security, and Developers. The main content area has a red header with a 'Quick search' bar and user controls. Below the header, a message states that activation of the DEPAOUI configuration for these versions does not take place automatically and that the MPS Monitor Portal must be accessed after the explorer has been correctly registered and configuration has been manually activated. A 'Customer Code' field contains the value '6XV2BAUFUY'. Two columns of product versions are listed, each with an 'Installation guide' link. The first column lists versions requiring only a customer code, and the second column lists versions requiring both a customer code and a license code. At the bottom right, under 'Available licenses', there is a 'Generate license' button highlighted with a red box and an arrow pointing to it. The text '0 results found' is displayed below the button.

► FIG 2.1.2-026 Installing the connector for all versions other than Windows-2

The newly generated license can be used immediately. The versions of the DCA connector that require both the customer code and the license code are as follows:

- Embedded Lexmark eXplorer: [click here to access the instructions](#).
- Embedded FutureSmart eXplorer: [click here to access the instructions](#)



→ eXplorer Mac Os v.3 RC1: [click here to access the instructions](#)

Each installation manual contains all the detailed information to install and configure the different versions of DCA connectors.

